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AI adoption and cross-border e-commerce in a fragmented and uncertain European Union

Abstract

Artificial intelligence (AI) is increasingly shaping economic activity across the European Union, yet its adoption remains highly uneven among member states. In parallel, cross-border e-commerce has become an important channel of digital integration, operating in an environment marked by economic fragmentation and heightened uncertainty. This project examines whether differences in AI adoption across EU countries are associated with cross-border e-commerce outcomes.

Using data from Eurostat and the European Central Bank, we analyse AI adoption among enterprises and its relationship with B2C and B2B e-commerce across EU countries. The results indicate a positive association between AI adoption and B2C e-commerce participation, with digitally more advanced economies achieving stronger outcomes in consumer-oriented online trade. This relationship remains qualitatively robust when a more restrictive measure of AI intensity is applied, while the link appears weaker for B2B e-commerce. Exchange rate developments provide macroeconomic context for these relationships.

Overall, the findings highlight AI diffusion as a factor shaping cross-border digital trade and suggest that uneven technological adoption may reinforce economic fragmentation within the EU, with important implications for digital and single market policies.